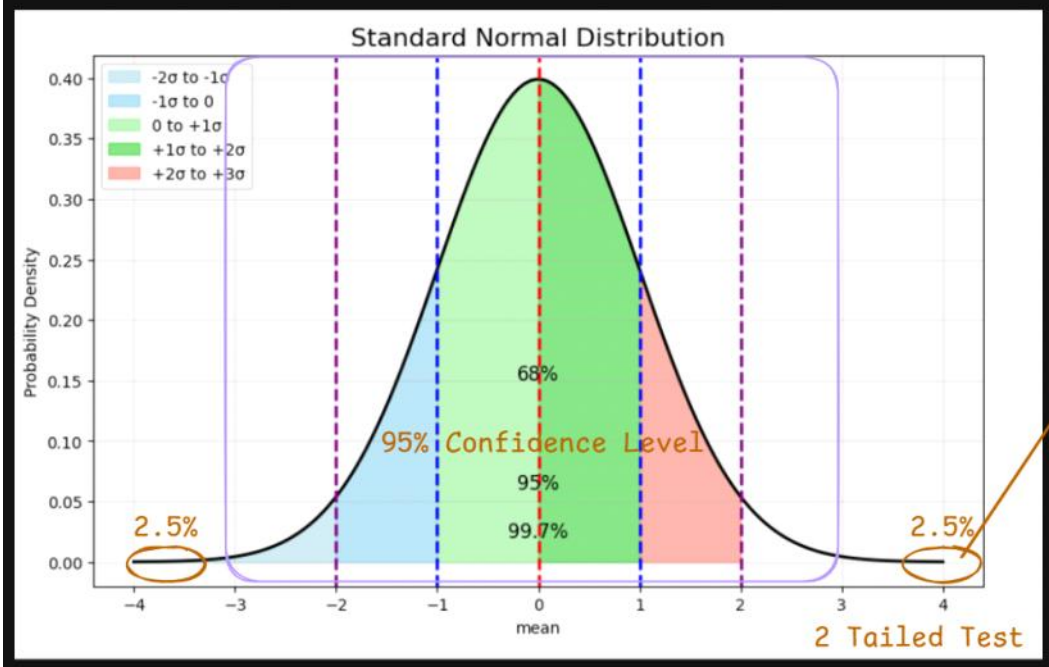
Hypothesis Testing - II

🎯 Session Objectives:

- ✅ Understand and articulate the steps involved in hypothesis testing.
- ✅ Formulate null and alternate hypotheses for various scenarios.
- ✅ Explain the role of normal and standard normal distributions in hypothesis testing.
- ✅ Define and interpret p-values and significance levels (alpha) in the context of hypothesis testing.
- ✅ Construct appropriate null and alternate hypotheses based on different practical situations.
- ✅ Solve problems on hypothesis testing
- ✅ Accept or reject hypothesis basis given statistics
- ✅ Learn how to design hypothesis
- ✅ Solve the hypothesis testing problems with defined steps..



alpha
5 % Significance Level

Reject H0 ~ Accept HA

H0 = Average Mean = 100
HA = Average Mean != 100

We have to perform 2-tailed Test. As no clear direction mentioned on Hypothesis

p_value < alpha (5%) : Significance level

p_value ~ 18% : H0 Accept
p_value ~ 3% < 5% : Ho Reject

H0 = Average Mean >= 100
HA = Average Mean < 100

We have to perform 1-tailed Test. As negative direction mentioned on Hypothesis

Significance levels (alpha) = 1 - Confidence Level

Confidence Level	Z-Score (Two-tailed)	% in Each Tail
80%	1.28	10%
85%	1.44	7.5%
90%	1.645	5%
92%	1.75	4%
95% alpha ~ 5% ~ .05	1.96	2.5%
96%	2.05	2%
97%	2.17	1.5%
98%	2.33	1%
99%	2.575	0.5%
99.5%	2.807	0.25%
99.9%	3.291	0.05%

https://math.arizona.edu/~rsims/ma464/standardnormaltable.pdf

Understanding Probability Density Function (PDF)

The PDF provides a way to describe continuous variable distributions. Unlike discrete variables, we cannot assign probability to a single point. Instead, the PDF gives us a density value f(x).

The probability of a variable falling within an interval [a, b] is found by integrating the PDF:

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

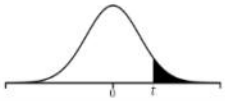
confidence Interval 95%

The Role of Normal Distribution

The normal distribution, or "bell curve," visualizes data where most values cluster around the mean. This distribution is symmetric, with data evenly spread around the center.

Why It Matters: When you gather a sample, your sample mean will likely fall somewhere along this curve. If close to the population mean, it will be near the center. If far away, it will be in the tails.

Critical Values for Student's t-Distribution.



df	Upper Tail Probability: $\Pr(T > t)$									
	0.2	0.1	0.05	0.04	0.03	0.025	0.02	0.01	0.005	0.0005
1	1.376	3.078	6.314	7.916	10.579	12.706	15.895	31.821	63.657	636.619
2	1.061	1.886	2.920	3.320	3.896	4.303	4.849	6.965	9.925	31.599
3	0.978	1.638	2.353	2.605	2.951	3.182	3.482	4.541	5.841	12.924
4	0.941	1.533	2.132	2.333	2.601	2.776	2.999	3.747	4.604	8.610
5	0.920	1.476	2.015	2.191	2.422	2.571	2.757	3.365	4.032	6.869
6	0.906	1.440	1.943	2.104	2.313	2.447	2.612	3.143	3.707	5.959
7	0.896	1.415	1.895	2.046	2.241	2.365	2.517	2.998	3.499	5.408
8	0.889	1.397	1.860	2.004	2.189	2.306	2.449	2.896	3.355	5.041
9	0.883	1.383	1.833	1.973	2.150	2.262	2.398	2.821	3.250	4.781
10	0.879	1.372	1.812	1.948	2.120	2.228	2.359	2.764	3.169	4.587

Z-Score

Tells us how many standard deviations a data point is away from the mean. For example, a z-score of 2 means the data point is 2 standard deviations from the mean.

The P-Value Explained

1

Definition

The p-value is the probability that the observed data (or something more extreme) would occur if the null hypothesis were true.

2

Interpretation

It tells us how likely it is to get our sample mean, given the null hypothesis. Lower p-values indicate stronger evidence against H_0.

3

Decision Rule

If p-value < α (significance level), reject H_0.
If p-value ≥ α, do not reject H_0.

Use a **z-score** when the **population standard deviation is known** and the **sample size is large (n ≥ 30)**, or when the population is normally distributed with known parameters. Use a **t-score** when the **population standard deviation is unknown** and must be estimated from the **sample standard deviation**, particularly when the **sample size is small (n < 30)**.

Calculate Z-Score

$$z = \frac{\bar{x} - \mu}{SE} = \frac{790 - 800}{7.5} \approx -1.33$$

The z-score formula standardizes the difference between sample and population means:

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$SE = \frac{\sigma}{\sqrt{n}}$$

← Standard deviation

← Number of samples

STANDARD NORMAL DISTRIBUTION: Table

Z	.00	.01	.02	.03
-3.9	.00005	.00005	.00004	.00004
-3.8	.00007	.00007	.00007	.00006
-3.7	.00011	.00010	.00010	.00010
-3.6	.00016	.00015	.00015	.00014
-3.5	.00023	.00022	.00022	.00021
-3.4	.00034	.00032	.00031	.00030
-3.3	.00048	.00047	.00045	.00043
-3.2	.00069	.00066	.00064	.00062
-3.1	.00097	.00094	.00090	.00087
-3.0	.00135	.00131	.00126	.00122
-2.9	.00187	.00181	.00175	.00169
-2.8	.00256	.00248	.00240	.00233
-2.7	.00347	.00336	.00326	.00317

```python
# Hypothesis Testing
from scipy.stats import norm
import numpy as np

# Flour Package Weight
population_mean = 1.0 # in kgs
sample_mean = 0.98 # in kgs
sample_size = 50 # n>=30 : Z-Score
std = 0.05 # is known to me

Standard_Error = std / np.sqrt(sample_size)
z_score = (sample_mean - population_mean) / Standard_Error
z_score
```

```
-2.8284271247461925
```

```python
H0 = "The average weight of packaged flour is 1 kg"
HA = "The average weight of packaged flour is not 1 kg" # != 1
# 2-Tailed Test [Alternative Hypothesis can be +ve or -ve]
p_value = 2 * norm.cdf(z_score)
p_value
```

```
0.00467773498104723
```

```python
alpha = 0.05 # Significance Level
if (p_value < alpha):
    print(f"Rejecting the Null Hypothesis H0: {H0}")
    print(f"Accepting the Alternative Hypothesis HA: {HA}")
else:
    print(f"Accepting the Null Hypothesis H0: {H0}")
```

```
Rejecting the Null Hypothesis H0: The average weight of packaged flour is 1 kg
Accepting the Alternative Hypothesis HA: The average weight of packaged flour is not 1 kg
```

This small p-value indicates strong evidence against the null hypothesis.

Note: `Flour Example Decision: Since our p-value (0.0047) is less than alpha (0.05), we reject the null hypothesis. This suggests the average flour weight is significantly different from 1 kg.`

We conclude that there is strong evidence to suggest that the average weight of the flour is not 1 kg, meaning the factory's claim may not be accurate."